

LCM & HCF – APTITUDE LEARNING NOTES

Complete concepts • methods • formulas • shortcuts • solved examples • practice

1. What are HCF and LCM?

HCF (Highest Common Factor), also called GCD (Greatest Common Divisor), is the greatest positive integer that divides all the given numbers exactly.

LCM (Least Common Multiple) is the smallest positive integer that is exactly divisible by all the given numbers.

Example: For 12 and 18, common factors are 1, 2, 3, 6 → HCF = 6. Common positive multiples include 36, 72, ... → LCM = 36.

2. Factors vs Multiples

Factor: A number that divides another number without a remainder. Factors of 12: 1, 2, 3, 4, 6, 12.

Multiple: A number obtained by multiplying a number by 1, 2, 3, ... Multiples of 12: 12, 24, 36, 48, ...

A positive integer has a finite number of positive factors but infinitely many positive multiples.

3. Prime Factorization Method

Write each number as a product of prime numbers. For HCF, take the **common primes with the smallest powers**. For LCM, take **every prime appearing with the largest power**.

Example: $72 = 2^3 \times 3^2$ and $120 = 2^3 \times 3 \times 5$.

$\text{HCF} = 2^3 \times 3 = 24$. $\text{LCM} = 2^3 \times 3^2 \times 5 = 360$.

Memory trick: **HCF** → minimum powers; **LCM** → maximum powers.

4. Listing Method

For small numbers, list factors or multiples.

Example: HCF of 18 and 24. Factors of 18 = 1, 2, 3, 6, 9, 18; factors of 24 = 1, 2, 3, 4, 6, 8, 12, 24 → HCF = 6.

Example: LCM of 6 and 8. Multiples of 6: 6, 12, 18, 24; multiples of 8: 8, 16, 24 → LCM = 24.

This method is easy but becomes slow for large numbers.

5. Division / Euclidean Algorithm for HCF

For two positive integers a and b ($a > b$): divide a by b. Replace the pair by b and the remainder. Continue until the remainder becomes 0. The last non-zero remainder is the HCF.

Example: HCF(252, 105): $252 = 105 \times 2 + 42$; $105 = 42 \times 2 + 21$; $42 = 21 \times 2 + 0$ → HCF = 21.

Formula: $\text{gcd}(a, b) = \text{gcd}(b, a \bmod b)$.

6. HCF × LCM Relationship

For two positive integers a and b:

$$\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$$

Example: If HCF = 12, LCM = 420, and one number is 60, the other = $(12 \times 420) / 60 = 84$.

Important: This direct product relationship is guaranteed for **two** numbers. Do not blindly apply it to three or more numbers.

7. HCF / LCM of Fractions

For positive fractions in reduced form:

HCF of fractions = HCF of numerators / LCM of denominators.

LCM of fractions = LCM of numerators / HCF of denominators.

Example: HCF of $\frac{2}{3}$ and $\frac{4}{9}$ = $\text{HCF}(2,4)/\text{LCM}(3,9)$ = $\frac{2}{9}$.

8. HCF / LCM of Decimals

Convert the decimals into integers by multiplying all by the same power of 10. Find the HCF/LCM of the resulting integers, then scale back.

Example: HCF of 1.2 and 1.8. Multiply by 10 $\rightarrow$ 12 and 18. HCF=6. Divide by 10 $\rightarrow$ **0.6**.

9. HCF of More Than Two Numbers

Find the HCF step-by-step: $\text{HCF}(a,b,c) = \text{HCF}(\text{HCF}(a,b),c)$.

Example: HCF(24, 36, 60): $\text{HCF}(24,36)=12$; $\text{HCF}(12,60)=12 \rightarrow$ answer **12**.

10. LCM of More Than Two Numbers

Similarly, $\text{LCM}(a,b,c) = \text{LCM}(\text{LCM}(a,b),c)$. Or use prime factorization and take maximum exponents.

Example: $\text{LCM}(12,15,18) = 2^2 \times 3^2 \times 5 = \mathbf{180}$.

11. Same Remainder Problems

If a divisor leaves the **same remainder** when dividing several numbers, it must divide their pairwise differences.

Example: Find the greatest number that divides 43, 91 and 139 leaving the same remainder. Differences: $91-43=48$ and $139-91=48$. $\text{HCF}(48,48)=48$.

Shortcut: Greatest divisor = HCF of all pairwise differences.

12. Same Remainder and Given Remainder

If numbers $N_1, N_2, \dots$ leave the same remainder r when divided by d , then d divides each $N_i - r$. If r is known, subtract r from each number and find the HCF.

Example: Find the greatest divisor that leaves remainder 5 when dividing 65, 95 and 125. Subtract 5 $\rightarrow$ 60, 90, 120. $\text{HCF} = 30$.

13. Least Number Divisible by Given Numbers

When a question asks for the smallest positive number exactly divisible by several numbers, find their LCM.

Example: Smallest number divisible by 8, 12 and 15: $8=2^3$, $12=2^2 \times 3$, $15=3 \times 5 \rightarrow \text{LCM}=2^3 \times 3 \times 5=120$.

14. Smallest Number to Add / Subtract

To make N divisible by d , find $N \bmod d$.

- **Smallest number to subtract** = remainder r .
- **Smallest positive number to add** = $d-r$, if $r \neq 0$; if $r=0$, add 0 for a non-strict interpretation.

Example: $9876 \div 11$ leaves remainder 8. Subtract 8 or add 3 to reach a multiple of 11.

15. Smallest Number Leaving a Given Remainder

If N leaves remainder r when divided by each of a, b, c , then $N-r$ is a common multiple of a, b, c . The smallest positive candidate is usually $\text{LCM}(a, b, c) + r$, subject to N being larger than r .

Example: Find the smallest number which leaves remainder 3 when divided by 4, 6 and 10. $\text{LCM}=60 \rightarrow N=60+3=63$.

16. Greatest Number Leaving a Given Remainder

If the same remainder r is left by all numbers, subtract r and take HCF.

Example: Greatest number dividing 83, 113, 143 and leaving remainder 3: $\text{HCF}(80, 110, 140)=10$.

17. Product and HCF / LCM

For two numbers a, b : if $\text{HCF}(a, b)=g$ and $\text{LCM}(a, b)=L$, then $ab=gL$.

If $a=gm$ and $b=gn$ with $\text{gcd}(m, n)=1$, then $\text{LCM}(a, b)=gm n$.

This representation is useful when HCF is given: divide both numbers by the HCF; the remaining parts are coprime.

18. Co-Prime Numbers

Two numbers are co-prime if their HCF is 1. They need not both be prime. Example: 8 and 15 are co-prime.

If a and b are co-prime, then $\text{LCM}(a, b)=ab$.

Important: Prime numbers such as 2 and 3 are co-prime, but composite pairs such as 8 and 15 can also be co-prime.

19. HCF and LCM Using Venn-Style Prime Factors

Think of prime factors as overlapping groups. The HCF contains only the overlap, with minimum powers. The LCM contains the complete union, with maximum powers.

Example: $36=2^2 \times 3^2$, $48=2^4 \times 3$. $HCF=2^2 \times 3=12$; $LCM=2^4 \times 3^2=144$.

20. Special HCF / LCM Patterns

Consecutive integers: HCF of n and $n+1$ is always 1.

$\gcd(n, n+1)=1$.

Consecutive even integers: HCF can be 2 or another common factor depending on the sequence.

Example: 8 and 10 $\rightarrow$ 2; 12 and 14 $\rightarrow$ 2.

One number divides another: HCF is the smaller number and LCM is the larger.

Example: 12 and 48 $\rightarrow$ $HCF=12$, $LCM=48$.

Co-prime pair: $HCF=1$ and $LCM=\text{product}$.

Example: 9 and 20 $\rightarrow$ $LCM=180$.

Same number: $HCF=LCM=\text{that number}$.

Example: 25 and 25 $\rightarrow$ both 25.

21. Word-Problem Translation Guide

Question wording	Usually use
Smallest number divisible by all	LCM
Least common multiple / first common multiple	LCM
Greatest number dividing all exactly	HCF
Greatest number leaving same remainder	HCF of differences
Smallest number leaving same remainder	$LCM + \text{remainder}$
Largest possible equal groups	HCF
Largest equal length / largest tile size	HCF
Events repeat together / bells ring together	LCM
Divide objects into maximum identical groups	HCF
Find another number when product/HCF/LCM are given	$HCF \times LCM = \text{product (two numbers)}$

22. Solved Aptitude Examples

Q1. Find HCF and LCM of 84 and 126.

Solution: $84=2^2 \times 3 \times 7$; $126=2 \times 3^2 \times 7$. $\text{HCF}=2 \times 3 \times 7=42$. $\text{LCM}=2^2 \times 3^2 \times 7=252$.

Q2. HCF is 18 and LCM is 540. If one number is 90, find the other.

Solution: Other = $(18 \times 540)/90 = 108$.

Q3. Find HCF of 391 and 299.

Solution: $391=299 \times 1 + 92$; $299=92 \times 3 + 23$; $92=23 \times 4 \rightarrow \text{HCF}=23$.

Q4. Find LCM of 24, 36 and 54.

Solution: $24=2^3 \times 3$; $36=2^2 \times 3^2$; $54=2 \times 3^3$. $\text{LCM}=2^3 \times 3^3=216$.

Q5. Find the greatest number that divides 65, 117 and 169 leaving the same remainder.

Solution: Differences: 52, 52, 104. $\text{HCF}=52 \rightarrow 52$.

Q6. Find the smallest number that leaves remainder 4 when divided by 6, 8 and 10.

Solution: $\text{LCM}(6, 8, 10)=120$. Therefore $N=120+4=124$.

Q7. Three bells ring every 12, 18 and 30 minutes. If they ring together now, after how long will they ring together again?

Solution: $\text{LCM}(12, 18, 30)=2^2 \times 3^2 \times 5=180$ minutes = 3 hours.

Q8. 144 chocolates and 96 biscuits must be packed into the maximum number of identical packets with no item left. Find packets.

Solution: Maximum packets = $\text{HCF}(144, 96)=48$. Each packet gets 3 chocolates and 2 biscuits.

Q9. Find the smallest number to add to 754 to make it divisible by 18.

Solution: $754 \div 18$ leaves 16. Add $18-16=2$, giving 756.

Q10. Find the smallest number to subtract from 1000 so that it is divisible by 24.

Solution: $1000 \bmod 24=16$. Subtract 16 $\rightarrow 984$.

Q11. Find LCM of two co-prime numbers 14 and 25.

Solution: Since $\text{HCF}=1$, $\text{LCM}=14 \times 25=350$.

Q12. If HCF of two numbers is 15 and their product is 13,500, find their LCM.

Solution: $\text{LCM}=13,500/15=900$.

23. Practice Questions

1. Find the HCF of 96 and 144.
2. Find the LCM of 24, 36 and 54.
3. Find the HCF of 391 and 299 using the Euclidean algorithm.
4. Find the LCM of 18, 24 and 30.
5. Two numbers have HCF 12 and LCM 420. If one number is 60, find the other.
6. Find the greatest number that divides 43, 91 and 139 leaving the same remainder.
7. Find the greatest number that divides 65, 95 and 125 leaving remainder 5.
8. Find the smallest number divisible by 16, 20 and 24.
9. Find the smallest number that leaves remainder 7 when divided by 12, 15 and 20.
10. Find the smallest number to add to 9876 to make it divisible by 11.

11. Find the smallest number to subtract from 1000 to make it divisible by 18.
12. Three lights flash every 8, 12 and 15 seconds. When will they flash together again?
13. Find the HCF and LCM of 72 and 120.
14. Find the HCF of 24, 36 and 60.
15. Find the LCM of 12, 15 and 18.
16. 180 students and 252 students are to be divided into maximum identical groups. Find the number of groups.
17. Find HCF and LCM of $\frac{2}{3}$ and $\frac{4}{9}$.
18. Find the HCF of 1.2 and 1.8.
19. Two co-prime numbers have product 391. If one is 17, find the other and their LCM.
20. If $\text{HCF}=16$, $\text{LCM}=240$ and one number=48, find the other.

24. Answer Key

1) 48 • 2) 216 • 3) 23 • 4) 360 • 5) 84 • 6) 48 • 7) 30 • 8) 240 • 9) 67 • 10) 3 • 11) 10 • 12) 120 seconds • 13) $\text{HCF}=24$, $\text{LCM}=360$ • 14) 12 • 15) 180 • 16) 36 • 17) $\text{HCF}=\frac{2}{9}$, $\text{LCM}=\frac{4}{3}$ • 18) 0.6 • 19) 23; $\text{LCM}=391$ • 20) 80

25. Quick Revision Formula Sheet

- **HCF:** common prime factors with minimum exponents.
- **LCM:** all prime factors with maximum exponents.
- **Two numbers:** $\text{HCF} \times \text{LCM} = \text{product of the numbers}$.
- **Euclidean:** $\text{gcd}(a,b)=\text{gcd}(b,a \bmod b)$.
- **Same remainder:** greatest divisor = HCF of pairwise differences.
- **Known remainder r:** subtract r first, then take HCF.
- **Smallest common multiple:** LCM.
- **Smallest number leaving remainder r:** generally $\text{LCM} + r$.
- **Largest equal groups:** HCF.
- **Two co-prime numbers:** $\text{HCF}=1$ and $\text{LCM}=\text{product}$.
- **One number divides another:** $\text{HCF}=\text{smaller}$; $\text{LCM}=\text{larger}$.
- **Fractions:** $\text{HCF} = \frac{\text{HCF}(\text{numerators})}{\text{LCM}(\text{denominators})}$; $\text{LCM} = \frac{\text{LCM}(\text{numerators})}{\text{HCF}(\text{denominators})}$, for positive fractions in reduced form.
- **Decimals:** scale to integers, solve, then scale back.

26. Exam Strategy

Step 1: Identify the wording: greatest divisor/group/length → HCF; smallest common/meeting/repeating → LCM.

Step 2: For small numbers, listing can be fastest. For larger numbers, prime factorization or Euclidean algorithm is safer.

Step 3: For same-remainder questions, immediately write the differences.

Step 4: Check the answer against the wording: HCF should divide every original number; LCM should be divisible by every original number.

Step 5: Practice timed questions. First aim for accuracy, then speed.